

Game Theory

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- Game is defined as an activity among two or more persons as per set of rule at the end of which each persons get some benefits or bear loss.
- The term 'game' represents a competition or conflict between two or more parties. (known as players).
- Game theory is a branch of applied mathematics that deals with interactive decision-making in situations of competition or conflict.
- It analyzes strategies where the outcome for each participant depends on the choices of all.

For example, if you're negotiating a salary with your boss, you could use game theory to determine your bargaining power and identify potential outcomes that would benefit both parties.

* Basic Terminology / Key concepts / characteristics

- players: competitors or decision makers in a game
- Strategies: Set of alternative courses of action available to each player based on advance knowledge
- pure strategy: If the player selects the same strategy each time.
- Mixed strategy: If the player selects the strategy according to some fixed probability
- optimal strategy: A strategy by which a player can achieve the best pay off.
- saddle point: The point where $\text{Maximin} = \text{Minimax}$
- value of game: The expected outcome resulted from each play. It is denoted as V .

* Two persons zero sum game

- The game involving two players in which the gain made by one player is equal to the loss of another.

Types :

- 1) The game with Saddle point (pure strategy)
- 2) The game without saddle point (Mixed strategy)

1) The game with Saddle point (pure strategy)

- It requires the following steps:

Step-1 • Select the minimum element from each row and write them in Row minimum column. Also select the maximum value from that column & enclose it in $[]$.

It is called Row Maximin

Step-2 • Select the maximum element from each column and write them in column Maximum row. Also select the minimum value from that row and enclose it in $()$. It is called column Minimax.

Step-3 • Find out the elements that is same in $[]$ & $()$. If column Minimax = Row Maximin, then the game has saddle point and it is the value of the game & the corresponding strategies are called optimal strategy.

Note: i) The game is said to be fair if

$$\text{Maximin} = \text{Minimax} = 0 \quad (\text{i.e. value of game } (V) = 0)$$

ii) The game is said to be strictly determinable if

$$\text{Maximin} = \text{Minimax} \neq 0$$

Example:

Q.1. Solve the following two-person-zero sum game with the following 3×2 payoff matrix

Player A	Player B	
	B_1	B_2
A_1	9	2
A_2	8	6
A_3	6	4

Soln

We apply the maximin (minimax) principle to analyze the game

Player A	Player B		Row-min
	B_1	B_2	
A_1	9	2	2
A_2	8	6	[6] (Maximin)
A_3	6	4	4
Col-max	9	(6)	

Now, (Minimax)

• select the maximum from the minimum of rows
 $\therefore$ Row Maximin = 6

• select the minimum from the maximum of columns
 $\therefore$ Col Minimax = 6

Since, Maximin = Minimax, the saddle point exists

$\therefore$ The optimal strategy of A is A_2 & B is B_2

Value of game (V) = 6

Q.2. Solve the game problem with following payoff matrix

	B_1	B_2	B_3	B_4	Row-min
A_1	-5	2	0	7	-5
A_2	5	6	4	8	[4] $\rightarrow$ Maximin
A_3	4	0	2	-3	-3
C-max	5	6	(4)	8	

$\therefore$ Maximin = ^{minimax} minimax, saddle point exists

$\therefore$ The optimal strategy of A is A_2 & B is B_2

Value of game (V) = 4

Q.3 Find range of p & q that will make the payoff matrix given below

Player A	Player B			R-min
	B_1	B_2	B_3	
A_1	2	4	7	2
A_2	10	7	9	7 $\rightarrow$ Maximin
A_3	4	p	8	4
C-Max	10	7	8	

(Minimax)

Here, min value in R_2 is 7, so, $q \geq 7$ & max value in C_3 is 8, so, $q \leq 8$

$\therefore$ Range of $q = 7 \leq q \leq 8$

For p

min value in R_3 is 4, so $p \geq 4$ & max value in C_2 is 7, so, $p \leq 7$

$\therefore$ Range of $p = 4 \leq p \leq 7$

2, The game without saddle point (Mixed strategy)

$\rightarrow$ A mixed strategy game can be solved by following method

i, Algebraic method iv, Arithmetic method

ii, Graphical method

iii, Dominance method

i, Algebraic method

$\rightarrow$ consider the two person zero sum game with following

Payoff matrix		Player B	
		I	II
Player A	I	a_{11}	a_{12}
	II	a_{21}	a_{22}

The solution of the game is

$\Rightarrow$ A plays (P_1, P_2)

$\Rightarrow$ B plays (Q_1, Q_2)

$$P_1 = \frac{a_{22} - a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

$$Q_1 = \frac{a_{22} - a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

$$P_2 = 1 - P_1$$

$$Q_2 = 1 - Q_1$$

$$\therefore \text{Value of game } (V) = \frac{a_{11}a_{22} - a_{21}a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

Example:

Q1, Solve the following game whose payoff matrix is

		Player B	
		B ₁	B ₂
Player A	A ₁	3	-2
	A ₂	2	5

-2
 [2] → Maximin
 C-max (3) 5
 Minimax

Since, Maximin $\neq$ Minimax, so saddle point doesn't exist.

Now,

A play's (p_1, p_2) solving using Algebraic method

$$\begin{aligned}
 \bullet p_1 &= \frac{a_{22} - a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{5 - 2}{(3 + 5) - (-2 + 2)} = \frac{3}{8} \\
 \bullet p_2 &= 1 - p_1 = 1 - \frac{3}{8} = \frac{5}{8}
 \end{aligned}$$

B play's (q_1, q_2)

$$\begin{aligned}
 \bullet q_1 &= \frac{a_{22} - a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{5 - (-2)}{(3 + 5) - (-2 + 2)} = \frac{7}{8} \\
 \bullet q_2 &= 1 - q_1 = 1 - \frac{7}{8} = \frac{1}{8}
 \end{aligned}$$

$$\therefore \text{value of game (V)} = \frac{a_{11}q_1 + a_{12}q_2 - a_{21}q_1 - a_{22}q_2}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{3 \times \frac{7}{8} - 2 \times \frac{1}{8}}{8} = \frac{19}{8}$$

$$\therefore \text{optimum mixed strategy of A} = \left(\frac{3}{8}, \frac{5}{8} \right) \text{ \& B} = \left(\frac{7}{8}, \frac{1}{8} \right)$$

ii) Arithmetic method / short-cut method

		Player B	
		B ₁	B ₂
Player A	A ₁	3	-2
	A ₂	2	5

$|2 - 5| = 3 \therefore p_1 = \frac{3}{3+5} = \frac{3}{8}$
 $|3 - (-2)| = 5 \therefore p_2 = \frac{5}{3+5} = \frac{5}{8}$
 $|5 - (-2)| = 7 \quad |3 - 2| = 1$

$$\therefore q_1 = \frac{7}{7+1} = \frac{7}{8} \quad \therefore q_2 = \frac{1}{7+1} = \frac{1}{8}$$

$$\therefore \text{optimal strategy of A} = \left(\frac{3}{8}, \frac{5}{8} \right) \text{ \& B} = \left(\frac{7}{8}, \frac{1}{8} \right)$$

$$\therefore \text{value of game (V)} = 3 \times \frac{3}{8} + 2 \times \frac{5}{8} = \frac{9}{8} + \frac{10}{8} = \frac{19}{8}$$

Q.2. Find the value of game, given the following payoff matrix

	B ₁	B ₂	R-min
A ₁	4	-4	(-4) → Maximin
A ₂	-4	4	
C-max	(4)	4	
Minimax			

∴ Maximin ≠ Minimax, so saddle point doesn't exist

Now, using Arithmetic method

$$\begin{array}{cc}
 & \begin{matrix} B_1 & B_2 \end{matrix} \\
 \begin{matrix} A_1 \\ A_2 \end{matrix} & \begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix}
 \end{array}$$

$| -4 - 4 | = 8 \quad \therefore p_1 = \frac{8}{8+8} = \frac{1}{2}$
 $| 4 - (-4) | = 8 \quad \therefore p_2 = \frac{8}{8+8} = \frac{1}{2}$
 $| 4 - 4 | = 8 \quad \therefore q_1 = \frac{1}{2}$
 $| 4 - (-4) | = 8 \quad \therefore q_2 = \frac{1}{2}$

∴ optimal strategy of A = $(\frac{1}{2}, \frac{1}{2})$

optimal strategy of B = $(\frac{1}{2}, \frac{1}{2})$

Expected gain of player A

• Value of game (V) : $4 \times \frac{1}{2} + -4 \times \frac{1}{2} = 0$ (When player B will employ strategy B₁)

or, $-4 \times \frac{1}{2} + 4 \times \frac{1}{2} = 0$ (" " strategy B₂)

Expected loss of player B

• Value of game (V) : $4 \times \frac{1}{2} + -4 \times \frac{1}{2} = 0$ (When player A will employ strategy A₁)

or $-4 \times \frac{1}{2} + 4 \times \frac{1}{2} = 0$ (" " strategy A₂)

Q.3 In game of matching coins, Player A wins Rs. 8, if both coins are heads & Rs. 1 if both coins are tails. Player B wins Rs. 3, when coin do not match. Given the choice of being player A or B, which would you choose and what would be your strategy.

Note: Here, in payoff matrix we are talking about player A. So, the wins of player B, is the loss of player A.

(i.e. gain of player B = - gain of player A)

→ Player B gains +3, then player A must lose -3

Sol 2

Representing the given case in payoff matrix:

Player B
H T
Player A
H $\begin{bmatrix} 8 & -3 \end{bmatrix}$ $| -3 - 1 | = 4 \therefore P_1 = \frac{4}{15}$
T $\begin{bmatrix} -3 & 1 \end{bmatrix}$ $| 8 - (-3) | = 11 \therefore P_2 = \frac{11}{15}$

$| -3 - 1 | = 4$ $| 8 - (-3) | = 11$

$\therefore q_1 = \frac{4}{15}$

$q_2 = \frac{11}{15}$

$\therefore$ optimal strategy of A = $\left(\frac{4}{15}, \frac{11}{15} \right)$ & B = $\left(\frac{4}{15}, \frac{11}{15} \right)$

Value of game (V) = $8 \times \frac{4}{15} - 3 \times \frac{11}{15} = -\frac{1}{15}$

* Dominance Rule

→ It is used to reduce size of pay off matrix in game theory

Note: R - min → eliminate minimum row

C - max → eliminate maximum column

(i) Row Dominance

	C ₁	C ₂	C ₃
R ₁	4	6	-5
R ₂	5	6	-3

$\therefore R_1 \leq R_2$, delete R₁

(ii) Column Dominance

	C ₁	C ₂
R ₁	4	2
R ₂	0	-1
R ₃	-3	-3

$\therefore C_1 \geq C_2$, remove C₂

Note: If there exists a case that, we are not able to compare between two rows or column, then we have to find average of two rows/columns & compare with third one.

	B ₁	B ₂
A ₁	2	4
A ₂	5	0
A ₃	0	8
A ₄	2.5	4

Aug

$\therefore A_1 \leq A_4$, remove A₁

	B ₁	B ₂	B ₃	avg
A ₁	8	15	11	8
A ₂	1	-1	4	1.5

$\therefore B_1 \geq B_4$, remove B₁

Example:

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Q.1. Reduce the following game using the concept of dominance

	B_1	B_2	B_3	B_4
A_1	3	5	4	2
A_2	5	6	2	4
A_3	2	1	4	0
A_4	3	3	5	2

$\therefore R_1 \leq R_4 \rightarrow \text{Remove } R_3$

Step: 2

	B_1	B_2	B_3	B_4
A_1	3	5	4	2
A_2	5	6	2	4
A_4	3	3	5	2

$\therefore C_2 \geq C_1 \rightarrow \text{Remove } C_2$ & $C_1 \geq C_4 \rightarrow \text{Remove } C_4$

Step: 3

	B_3	B_4
A_1	4	2
A_2	2	4
A_3	5	2

$\therefore R_1 \leq R_3 \rightarrow \text{Remove } R_2$

Q.2. Reduce the following game using dominance method and solve it

	B_1	B_2	B_3	B_4
A_1	3	2	4	0
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

$\therefore R_1 \leq R_3 \rightarrow \text{remove } R_1$

Step-2:

	B_1	B_2	B_3	B_4
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

$\therefore C_1 \geq C_3 \rightarrow \text{remove } C_1$

Step-3:

	B ₂	B ₃	B ₄	
A ₂	4	2	4	
A ₃	2	4	0	
A ₄	4	0	8	
A ₅	3	2	4	Avg

	B ₂	B ₃	B ₄	Avg
A ₂	4	2	4	3
A ₃	2	4	0	2
A ₄	4	0	8	4

$\therefore C_2 \geq \text{Avg of } (C_3, C_4) \rightarrow \text{Remove } C_2$

$\therefore A_5 \leq A_2 \rightarrow \text{Remove } A_5$

A_5 can't be removed, Find Avg. column

Step-4:

	B ₂	B ₃	B ₄	Avg
A ₃	2	4	0	2
A ₄	4	0	8	4

	B ₃	B ₄	
A ₂	2	4	
A ₃	4	0	
A ₄	0	8	
Avg	2	4	

$\therefore C_1 \geq \text{Avg of } (C_2, C_3) \rightarrow \text{Remove } C_1$

$\therefore R_1 \leq \text{Avg of } (R_2, R_3) \rightarrow \text{Remove } R_1$

Step-5:

	B ₃	B ₄	
A ₃	4	0	
A ₄	0	8	

$|0-8| = 8 \quad |4-0| = 4$

$q_1 = \frac{2}{3} \quad q_2 = \frac{1}{3}$

$\therefore$ optimal strategy of A = $(0, 0, \frac{2}{3}, \frac{1}{3})$

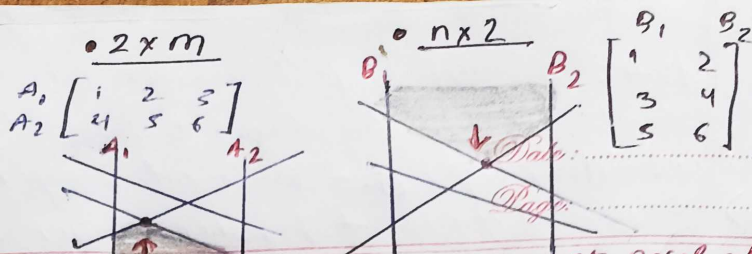
" " B = $(0, 0, \frac{2}{3}, \frac{1}{3})$

Value of game (V) = $4 \times \frac{2}{3} + 0 \times \frac{1}{3}$

= $\frac{8}{3}$ //

* Graphical Method

→ Useful for $2 \times m$ & $n \times 2$ order



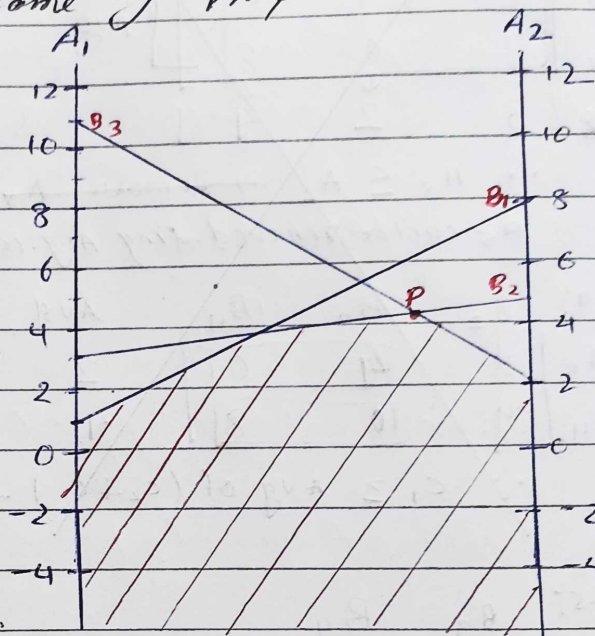
Q.1, Solve the following 2×3 game by Graphical method

	B_1	B_2	B_3
A_1	1	3	11
A_2	8	5	2

From the graph

The highest point (P) of the shaded region (lower envelope) is the intersecting point of B_2 & B_3 .

So, we consider only two strategies B_2 & B_3 for player B.



Thus, the solution to the original (2×3) game reduces to:

$$\begin{array}{cc}
 B_2 & B_3 \\
 A_1 & \begin{bmatrix} 3 & 11 \end{bmatrix} \\
 A_2 & \begin{bmatrix} 5 & 2 \end{bmatrix}
 \end{array}$$

$$|11 - 2| = 9 \quad |3 - 5| = 2$$

$$\therefore q_1 = 9/11 \quad \therefore q_2 = 2/11$$

$$\therefore \text{The optimal strategy of A} = \left(\frac{3}{11}, \frac{8}{11} \right)$$

"

$$B = \left(0, \frac{9}{11}, \frac{2}{11} \right)$$

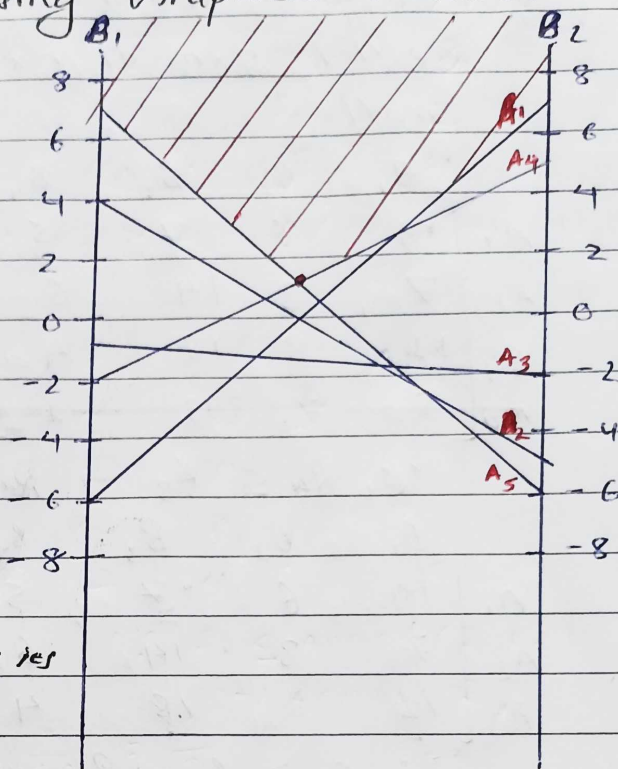
$$\therefore \text{Value of game } (V) = \frac{3 \times 3}{11} + \frac{5 \times 8}{11}$$

$$= \frac{9}{11} + \frac{40}{11}$$

$$= \frac{49}{11} \quad //$$

Q.2. Solve the following game using Graphical method.

	B_1	B_2
A_1	-6	7
A_2	4	-5
A_3	-1	-2
A_4	-2	5
A_5	7	-6



From the graph

The lowest point of the shaded region (upper envelope) is the intersecting point of A_4 & A_5

So, we consider only two strategies A_4 & A_5 for player A.

Thus.

$$A_4 \begin{bmatrix} -2 & 5 \end{bmatrix} \begin{matrix} |7-(-6)| = 13 \\ |-2-5| = 7 \end{matrix} \therefore p_1 = 13/20$$

$$A_5 \begin{bmatrix} 7 & -6 \end{bmatrix} \begin{matrix} |7-(-6)| = 13 \\ |-2-5| = 7 \end{matrix} \therefore p_2 = 7/20$$

$$|5-(-6)| = 11 \quad |-2-7| = 9$$

$$\therefore q_1 = \frac{11}{20} \quad \therefore q_2 = \frac{9}{20}$$

$$\therefore \text{Optimal strategy of A} = \left(0, 0, 0, \frac{13}{20}, \frac{7}{20} \right)$$

$$\text{Optimal " B} = \left(\frac{11}{20}, \frac{9}{20} \right)$$

$$\therefore \text{Value of game (V)} = -2 \times \frac{13}{20} + 7 \times \frac{7}{20}$$

$$= \frac{-26}{20} + \frac{49}{20}$$

$$= \frac{23}{20} \quad //$$

Q.3. Use the dominance rule to reduce the following game either $2 \times n$ or $n \times 2$ game & then solve by graphical method

	B_1	B_2	B_3	B_4
A_1	19	6	7	5
A_2	7	3	14	6
A_3	12	8	18	4
A_4	8	7	13	-1

$\therefore R_4 \leq R_3 \rightarrow \text{Remove } R_4$

	B_1	B_2	B_3	B_4
A_1	19	6	7	5
A_2	7	3	14	6
A_3	12	8	18	4

$\therefore C_1 \geq C_2 \rightarrow \text{Remove } C_1$ & $C_3 \geq C_4 \rightarrow \text{Remove } C_3$

The reduced matrix is

	B_2	B_4
A_1	6	5
A_2	3	6
A_3	8	4

Now, using graphical method to solve it
From graph

The lowest point from the shaded region is the intersecting point of A_1 & A_2 .

So, we consider only two strategies A_1 & A_2 for Player A.

Then,

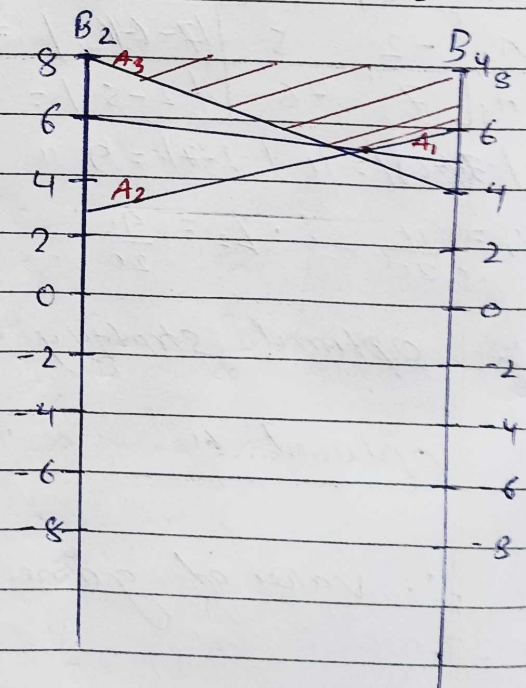
	B_2	B_4
A_1	6	5
A_2	3	6
	1	3

$|3-6| = 3 \therefore P_1 = \frac{3}{4}$
 $|6-5| = 1 \therefore P_2 = \frac{1}{4}$

$\therefore q_1 = \frac{1}{4} \quad q_2 = \frac{3}{4}$

$\therefore \text{Optimal strategy of A} = \left(\frac{3}{4}, \frac{1}{4}, 0, 0 \right)$
 $B = \left(0, \frac{1}{4}, 0, \frac{3}{4} \right)$

$\therefore \text{Value of game } (V) = 6 \times \frac{3}{4} + 3 \times \frac{1}{4} = 2\frac{1}{4}$



2023 Past Questions

Q. 11, a. Two types of food manufactures are competing for an increased market share. The payoff matrix describes the increase in market share of A and decrease share of B. Determine optimal strategies for both the manufacturers and the value of the game.

A	B			
	Give prizes	Decrease price	Maintain present strategy	Increase advertising
Give prizes	25	20	14	30
Decrease price	27	16	12	14
Maintain present strategy	35	8	15	19
Increase advertising	-2	8	13	0

Soln

Let the given payoff matrix be:

Step-1	B ₁	B ₂	B ₃	B ₄	Step-2	B ₁	B ₂	B ₃	B ₄
A ₁	25	20	14	30	A ₁	25	20	14	30
A ₂	27	16	12	14	A ₂	27	16	12	14
A ₃	35	8	15	19	A ₃	35	8	15	19
A ₄	-2	8	13	0					

$\therefore C_1 \geq C_2 \rightarrow \text{Remove } C_2$
 $C_4 \geq C_3 \rightarrow \text{Remove } C_3$
 $\therefore R_4 \leq R_3 \rightarrow \text{Remove } R_4$

Step-3	B ₂	B ₃
A ₁	20	14
A ₂	16	12
A ₃	8	15

$\therefore R_2 \leq R_1 \rightarrow \text{Remove } R_1$

Step-4	B ₂	B ₃
A ₂	16	12
A ₃	8	15

$|8-15| = 7 \therefore P_1 = 7/11$
 $|16-12| = 4 \therefore P_2 = 4/11$
 $\therefore \text{optimal strategy of A} = \left(0, \frac{7}{11}, \frac{4}{11}, 0\right)$
 $\therefore Q_1 = 3/11 \therefore Q_2 = 8/11$
 $B = \left(0, \frac{3}{11}, \frac{8}{11}, 0\right)$

$\therefore \text{The value of game (V)} = 16 \times \frac{7}{11} + 8 \times \frac{4}{11} = \frac{144}{11}$

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Q. 4 Reduce the following two person zero sum game to 2×2 order by dominance rule and obtain the optimal strategies for each player and the value of the game

Player A	Player B			
	B_1	B_2	B_3	B_4
A_1	3	2	4	0
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

Soln:

Since, $R_1 \leq R_3 \rightarrow$ Remove R_1

	B_1	B_2	B_3	B_4
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

Since, $C_1 \geq C_3 \rightarrow$ Remove C_1

	B_2	B_3	B_4	Avg		B_3	B_4
A_2	4	2	4	3	$\Rightarrow$	2	4
A_3	2	4	0	2		4	0
A_4	4	0	8	4		0	8
						2	4

$\therefore C_1 \geq \text{Avg of } (C_2, C_3) \rightarrow$ Remove C_1

Now, using Arithmetre method to solve the problem

	B_3	B_4	
A_2	4	0	8
A_4	0	8	4
	8	4	

$$\therefore P_1 = \frac{8}{12} = \frac{2}{3}$$

$$\therefore P_2 = \frac{4}{12} = \frac{1}{3}$$

$$Q_1 = \frac{2}{3} \quad Q_2 = \frac{1}{3}$$

$\therefore$ Optimal Strategy of A = $\left(\frac{2}{3}, \frac{1}{3}\right)$

"

$$B = \left(0, 0, \frac{2}{3}, \frac{1}{3}\right)$$

$$\therefore \text{Value of game (V)} = 4 \times \frac{2}{3} + 0 = \frac{8}{3}$$